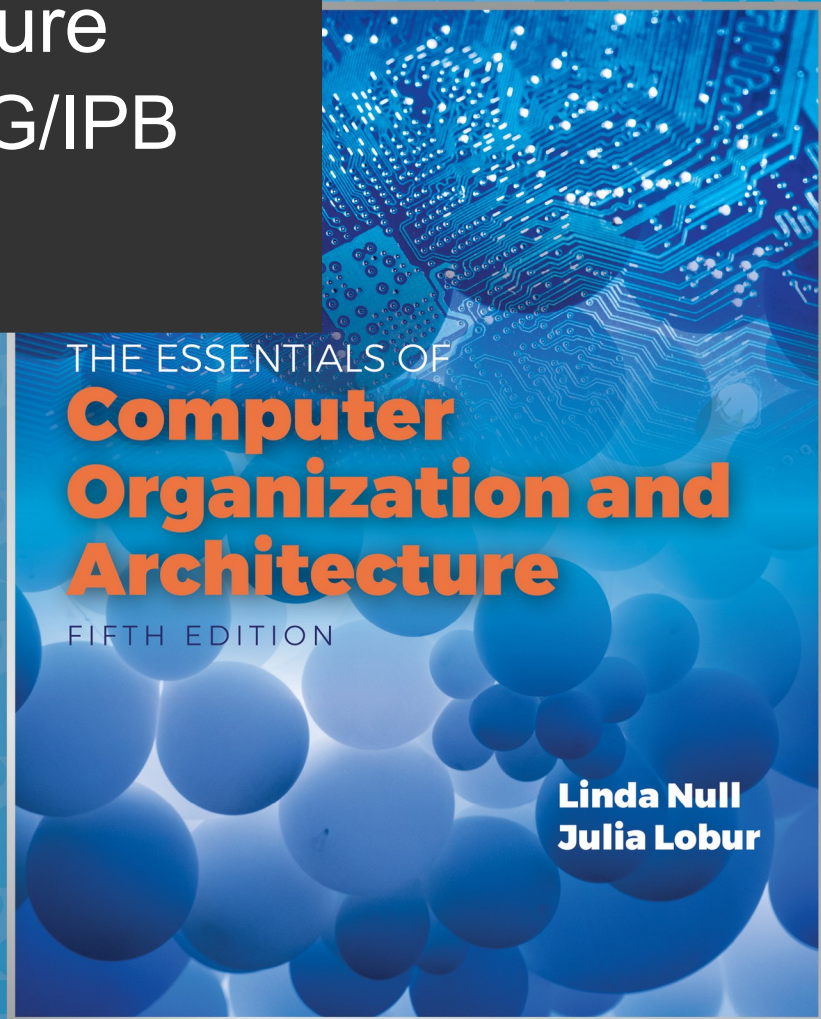


Computer Architecture Informatics Eng., ESTiG/IPB 2025/2026

José Rufino

(based on Chapter 1 of the book “The Essentials of Computer Organization and Architecture” of Linda Null & Julia Lobur)

Unit 1: Introduction



Objectives

- know the difference between **computer organization** and **computer architecture**
- understand the **units of measure** common to computer systems
- recognize the **evolution of computers**
- understand the computer as a **layered system**
- be able to explain the **von Neumann architecture** and the role of basic computer components

1.1 Overview



Why study computer organization and architecture?

- design better programs, including system software such as compilers, operating systems, and device drivers
- optimize program behavior
- evaluate (benchmark) computer system performance
- understand time, space, and economical trade offs

1.1 Overview



- **computer organization**
 - includes all **physical** aspects of computer systems
 - circuit design, control signals, memory types
 - *How does a computer work?*
- **computer architecture**
 - relates to the **logical** aspects of system implementation as seen by the programmer
 - instruction sets, instruction formats, data types, addressing modes, number and type of registers
 - *How to design a computer?*

1.1 Overview



- **computer architecture**
 - the combination of **Hardware Components** with the **Instruction Set Architecture (ISA)**
 - the ISA defines the **interface** between
 - ... the **software** that runs in the computer,
 - ... and the **hardware** that executes it
 - *the ISA allows to interact with the machine*

1.1 Overview



- in reality, there is no clear distinction between the issues related to computer **organization** and the ones that are relevant to computer **architecture**...
 - not entirely clear where **SW** ends and **HW** begins !
 - example:
 - to solve a **problem**, an **algorithm** is conceived
 - the coding of an algorithm produces a **program**
 - to execute the program, other programs (the OS) are necessary to create appropriate conditions
 - the HW itself is an implementation of algorithms

1.1 Overview



- “computers are implementations of algorithms which, in turn, execute other algorithms”
- **Principle of Equivalence of Hardware and Software:**
 - ***anything that can be done with software can also be done with hardware, and also the other way around****
 - * assuming speed is not a concern*
 - a generic use computer may be programmed to perform any task (flexibility at the cost of speed)
 - a specialized computer (e.g., embedded system) favors speed and simplicity, but is less flexible

1.2 Units of Measure

- units of measure commonly used in the area of information and communication technologies

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	μ	1 millionth = 10^{-6}	2^{-20}
Giga	G	1 billion = 10^9	2^{30}	Nano	n	1 billionth = 10^{-9}	2^{-30}
Tera	T	1 trillion = 10^{12}	2^{40}	Pico	p	1 trillionth = 10^{-12}	2^{-40}
Peta	P	1 quadrillion = 10^{15}	2^{50}	Femto	f	1 quadrillionth = 10^{-15}	2^{-50}
Exa	E	1 quintillion = 10^{18}	2^{60}	Atto	a	1 quintillionth = 10^{-18}	2^{-60}
Zetta	Z	1 sextillion = 10^{21}	2^{70}	Zepto	z	1 sextillionth = 10^{-21}	2^{-70}
Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

- powers of 2 decrease / increase faster !

n	m	$(2^n - 10^m) / 10^m$
10	3	2,40%
20	6	4,86%
30	9	7,37%
40	12	9,95%
50	15	12,59%

n	m	$(2^n - 10^m) / 10^m$
-10	-3	-2,34%
-20	-6	-4,63%
-30	-9	-6,87%
-40	-12	-9,05%
-50	-15	-11,18%

1.2 Units of Measure



- when do we use **base 10** or **base 2** ?
- **base 10**:
 - power (watts - W), current (ampere - A), voltage (volts - V), frequency (hertz - Hz (cycles/s))
 - transfer rates or bandwidth (bits per second - bps)
 - total capacity of hard disks and DVDs (bytes - B)
 - time (seconds - s) and distance (meters - m)
- **base 2**:
 - total capacity of RAM (bytes - B)
 - size of the sector for hard disks (e.g., 4KB)
 - total capacity of CDs (bytes - B)

1.2 Units of Measure



- **base 10**
 - **SI** (International System of Units) Prefixes

Prefix	Symbol	1000^m	10^n	Decimal	Short scale	Long scale	Since ^[1]
yotta	Y	1000^8	10^{24}	1 000 000 000 000 000 000 000 000	Septillion	Quadrillion	1991
zetta	Z	1000^7	10^{21}	1 000 000 000 000 000 000 000	Sextillion	Trilliard	1991
exa	E	1000^6	10^{18}	1 000 000 000 000 000 000	Quintillion	Trillion	1975
peta	P	1000^5	10^{15}	1 000 000 000 000 000	Quadrillion	Billiard	1975
tera	T	1000^4	10^{12}	1 000 000 000 000	Trillion	Billion	1960
giga	G	1000^3	10^9	1 000 000 000	Billion	Milliard	1960
mega	M	1000^2	10^6	1 000 000	Million		1960
kilo	k	1000^1	10^3	1 000	Thousand		1795
hecto	h	$1000^{2/3}$	10^2	100	Hundred		1795
deca	da	$1000^{1/3}$	10^1	10	Ten		1795
		1000^0	10^0	1	One		

for
BIG
amounts

1.2 Units of Measure



- **base 10**
 - **SI** (International System of Units) Prefixes

Prefix	Symbol	1000^m	10^n	Decimal	Short scale	Long scale	Since ^[1]
		1000^0	10^0	1	One		
deci	d	$1000^{-1/3}$	10^{-1}	0.1	Tenth		1795
centi	c	$1000^{-2/3}$	10^{-2}	0.01	Hundredth		1795
milli	m	1000^{-1}	10^{-3}	0.001	Thousandth		1795
micro	μ	1000^{-2}	10^{-6}	0.000 001	Millionth		1960 ^[2]
nano	n	1000^{-3}	10^{-9}	0.000 000 001	Billionth	Milliardth	1960
pico	p	1000^{-4}	10^{-12}	0.000 000 000 001	Trillionth	Billionth	1960
femto	f	1000^{-5}	10^{-15}	0.000 000 000 000 001	Quadrillionth	Billiardth	1964
atto	a	1000^{-6}	10^{-18}	0.000 000 000 000 000 001	Quintillionth	Trillionth	1964
zepto	z	1000^{-7}	10^{-21}	0.000 000 000 000 000 000 001	Sextillionth	Trilliardth	1991
yocto	y	1000^{-8}	10^{-24}	0.000 000 000 000 000 000 000 001	Septillionth	Quadrillionth	1991

for
small
amounts

1.2 Units of Measure

checkpoint: exercises
1.2, 1,2*, 1.3

- **base 2**
 - **IEC** (Int. Electrotechnical Commission) Prefixes

IEC prefix		Representations				Customary prefix	
Name	Symbol	Base 2	Base 1024	Value	Base 10	Name	Symbol
kibi	Ki	2^{10}	1024^1	1 024	$\sim 1.02 \times 10^3$	kilo	k, K
mebi	Mi	2^{20}	1024^2	1 048 576	$\sim 1.05 \times 10^6$	mega	M
gibi	Gi	2^{30}	1024^3	1 073 741 824	$\sim 1.07 \times 10^9$	giga	G
tebi	Ti	2^{40}	1024^4	1 099 511 627 776	$\sim 1.10 \times 10^{12}$	tera	T
pebi	Pi	2^{50}	1024^5	1 125 899 906 842 624	$\sim 1.13 \times 10^{15}$	peta	P
exbi	Ei	2^{60}	1024^6	1 152 921 504 606 846 976	$\sim 1.15 \times 10^{18}$	exa	E
zebi	Zi	2^{70}	1024^7	1 180 591 620 717 411 303 424	$\sim 1.18 \times 10^{21}$	zetta	Z
yobi	Yi	2^{80}	1024^8	1 208 925 819 614 629 174 706 176	$\sim 1.21 \times 10^{24}$	yotta	Y

for *BIG* amounts

1.3 Computer Components



- at the most basic level, a computer consists of:
 - a **processor** to interpret and execute programs
 - a **memory** to store both data and programs
 - a mechanism for transferring data with the outside world
- these components will be later studied in detail
- but for now, deciphering a sale announcement (see next slide) is a pretext for a first contact with them
- you should also watch the proposed videos (see slide ?)

1.3.1 An Example System

SSD

Quad Core

SDRAM

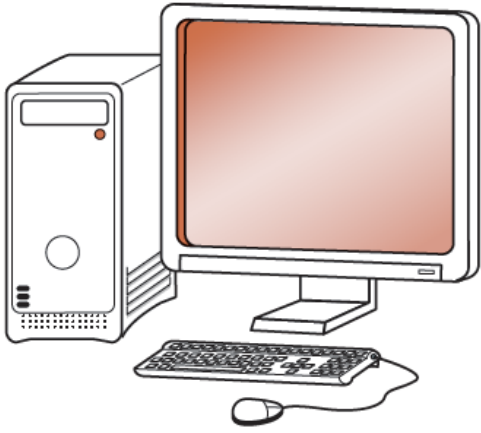
802.11

USB

PCIe

L1 cache

FOR SALE: OBSOLETE COMPUTER – CHEAP! CHEAP! CHEAP!



- Compact computer
- Intel i7 Quad Core, 4.20 GHz
- 2400MHz 32GB DDR4 SDRAM
- 128KB L1 cache, 2MB L2 cache
- Dual storage (7200RPM SATA 1TB HDD, 128GB SSD)
- Wireless 802.11 + Bluetooth 4.0
- 7-in-1 card reader
- 10 USB ports, 1 serial port, 4 PCI expansion slots (1 PCI, 1 PCI x 16, 2 PCI x 1), HDMI
- 24" widescreen LCD monitor, 16:10 aspect ratio, 1920 x 1200 WUXGA, 300 cd/m², active matrix, 1000:1 (static), 8ms, 24-bit color (16.7 million colors), VGA/DVI input, 2 USB ports
- 16x CD/DVD +/- RW drive
- 1GB PCIe video card
- PCIe sound card
- Gigabit Ethernet

What does it all mean??

1.3.1 An Example System

The processor is the “brain” of the system. It executes program instructions. This one is an Intel i7 Quad Core (4 cores) with a clock frequency of 4.2GHz.

Currently:

4/8/16/32/64/128... cores, 3.x/4.x/5.x GHz



- Compact computer
- Intel i7 Quad Core, 4.20 GHz
 - 2400MHz 32GB DDR4 SDRAM
 - 12MB L1 cache, 2MB L2 cache
 - Dual storage (7200RPM SATA 1TB HDD, 128GB SSD)
 - Wireless 802.11 + Bluetooth 4.0
 - 7-in-1 card reader
 - 10 USB ports, 1 serial port, 4 PCI expansion slots (1 PCI, 1 PCI x 16, 2 PCI x 1), HDMI
 - 24" widescreen LCD monitor, 16:10 aspect ratio, 1920 x 1200 WUXGA, 300 cd/m², active matrix, 1000:1 (static), 8ms, 24-bit color (16.7 million colors), VGA/DVI

A system bus moves data within the computer. The faster the bus the better. This one connects CPU to RAM and runs at 2400MHz.

Currently: DDR4 1600 ... 3200 MHz, DDR5 4000 ... 6400 MHz

1.3.1 An Example System



- the operation of the processor is regulated by a **clock**, which ensures that the components of the processor itself (or others that interact with it) are all synchronized
- a clock sends *electrical pulses*, simultaneously, to the various components connected to it; establishes opportune moments for the start of operations, allows the measurement of the duration of operations and the verification of their conclusion
- **cycle time (or period) = reciprocal of frequency**
 - for a frequency of 3.06 GHz = 3.06 G cycles/s = 3.06×10^9 cycles/s, the period is $\approx 0.326 \times 10^{-9}$ s/cycle = 0.326 ns/cycle

1.3.1 An Example System



- instruction duration is often measured in clock cycles
 - simpler (complex) instructions, take less (more) cycles
- CPUs of the same architecture generation, but higher frequency tend to run programs faster...
 - but be careful: the number of instructions completed per clock cycle is a more reliable metric of comparison
 - example: AMD Athlon *vs* Intel Pentium 4
- in addition to CPU frequency, many other factors have a decisive influence on performance:
 - bus speeds, amount of cache and main memory, etc.

1.3.1 An Example System

This system has 32GB of Synchronous Dynamic RAM (SDRAM) with Quad Data Rate transfers (DDR4) ...

Currently: DIMMs of 8/16/32/.../256GB/512GB, total of 8/16/32/64/128/256/512/.../6TB (per CPU socket)

... and two levels of cache memory: the level 1 (L1) cache is smaller and faster than the L2 cache.

Nowadays: 4/6/8/12/.../256/512 MB/1 GB L3, in addition to L1 and L2; CPUs with GPU Iris Pro: 128MB of L4 eDRAM cache

COMPUTER – CHEAP! CHEAP! CHEAP!

Compact computer

Intel® Quad Core, 4.20 GHz

2400MHz 32GB DDR4 SDRAM

• 128KB L1 cache, 2MB L2 cache

• Dual storage (7200RPM SATA 1TB HDD, 128GB SSD)

• Wireless 802.11 + Bluetooth 4.0

• 7-in-1 card reader

• 10 USB ports, 1 serial port, 4 PCI expansion slots
(1 PCI, 1 PCI x 16, 2 PCI x 1), HDMI

• 24" widescreen LCD monitor, 16:10 aspect ratio,
1920 x 1200 WUXGA, 300 cd/m², active matrix, 1000:1
(static), 8ms, 24-bit color (16.7 million colors), VGA/DVI

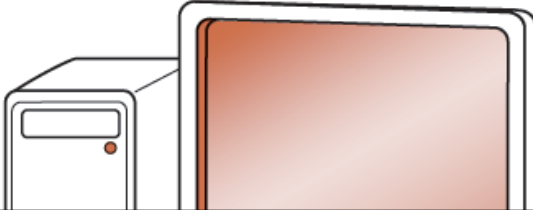
1.3.1 An Example System



- computers with more main memory (RAM) and cache can run larger programs, with more speed
- **RAM** is an acronym for *random access memory*
 - *random access* means that memory contents can be accessed **directly** by its location (address)
- **Cache** is a type of temporary memory that can be accessed faster than RAM (but it is smaller).
 - it sits “between” the CPU and the RAM
 - keeps copies of RAM blocks recently accessed

1.3.1 An Example System

Hard disk capacity determines the amount of data and the size of programs that can be stored.



OBsolete COMPUTER – CHEAP! CHEAP! CHEAP!

- Compact computer
- Intel i7 Quad Core, 4.20 GHz
- 2400MHz 32GB DDR4 SDRAM
- 128KB L1 cache, 2MB L2 cache
- **Dual storage (7200RPM SATA 1TB HDD, 128GB SSD)**
- Wireless 802.11n + Bluetooth 4.0
- 7-in-1 card reader

PCI expansion slots
HDMI
16:10 aspect ratio,
1920x1080, active matrix, 1000:1
contrast, 16.7 million colors, VGA/DVI

• 1GB PCIe video card

Magnetic disk of 1TB of capacity. 7200 RPM is the rotational speed of its platters (in general, the faster a disk rotates, the faster it can deliver data to RAM).

Solid State Disk (SSD) of only 128 GB of capacity. But much faster than the magnetic disk.

**Currently: disks of 8/16/24/32/36TB;
SSDs of 120,250,500 GB and 1/2/4(*)/
8(*)/.../64(*) TB ((*)servers)**

1.3.1 An Example System

SATA (Serial Advanced Technology Attachment): a standard for how hard disks interact with the system

Currently: SATA 3 (6Gb/s, 0.6GB/s), SAS 3 (12Gb/s, 1.2GB/s), SAS 4 (24Gb/s, 2.4GB/s), NVMe PCIe3 (2.4GB/s), NVMe PCIe4 (7 GB/s), NVMe PCIe5 (14 GB/s)

Nowadays: BD-ROM (25 / 50 GB); other formats may reach TBs ...

A DVD-ROM can store about 4.7GB (SL) / 8.7GB (DL) of data. This drive supports rewritable (RW) media; reading speed is 16x (wiring is usually less).

COMPLETE COMPUTER - CHEAP! CHEAP! CHEAP!

- Compact computer
- Intel i7 Quad Core, 4.20 GHz
- 2400MHz 32GB DDR4 SDRAM
- 128KB L1 cache, 2MB L2 cache
- Dual storage (7200RPM SATA 1TB HDD, 128GB SSD)
- Wireless 802.11 + Bluetooth 4.0
- 7-in-1 card reader
- 10 USB ports, 1 serial port, 4 PCI expansion slots (1 PCI, 1 PCI x 16, 2 PCI x 1), HDMI
- 24" widescreen LCD monitor, 16:10 aspect ratio, 1920 x 1200 WUXGA, 300 cd/m², active matrix, 1000:1 (static), 8ms, 24-bit color (16.7 million colors), VGA/DVI input, 2 USB ports
- 16x CD/DVD +/- RW drive
- 1GB PCIe video card

1.3.1 An Example System

Ports allow movement of data between a system and its external devices.

SD (Secure Digital) card reader supporting 7 formats.

This system has 10 USB ports, 1 serial port and 5 PCI slots.

FOR SALE: OBSOLETE COMPUTER – CHEAP! CHEAP! CHEAP!

- Compact computer
- Intel i7 Quad Core, 4.20 GHz
- 2400MHz 32GB DDR4 SDRAM
- 128KB L1 cache, 2MB L2 cache
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- Wireless 802.11 + Bluetooth 4.0
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- 16x CD/DVD +/- RW drive
- 1GB PCIe video card

Currently: USB2 (480 Mbps), USB3.0/3.1 (5/10 Gbps), Thunderbolt 2 (20 Gbps), Thunderbolt 3 (40 Gbps), Thunderbolt 5 (80 Gbps)

1.3.1 An Example System



- **serial** ports send data as a **sequence of pulses** along **one or two data lines**.
- **parallel** ports send data as a **single pulse** along at least **eight data lines (simultaneously)**
- **USB** (Universal Serial Bus) is an intelligent serial interface that is self-configuring (“plug and play”)

1.3.1 An Example System

System buses can be augmented by dedicated I/O buses. PCI[e] (*Peripheral Component Interconnect [express]*), is one such bus.

**PCI (1993), PCI-X (2004):
parallel operation;
PCIe 2.0 (500 MB/s x1):
serial operation**

This system has two PCIe devices:
a video card, and a sound card.

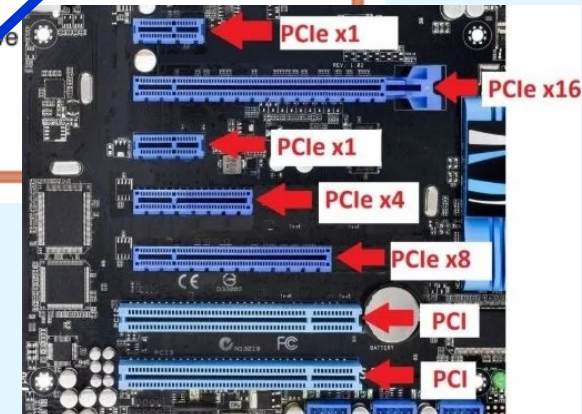
- Compact computer
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- Dual storage (7200RPM SATA 1TB HDD, 128GB SSD)

1920 x 1200 WUXGA, 300 cd/m², active matrix, 1000:1 (static), 8ms, 24-bit color (16.7 million colors), VGA/DVI input, 2 USB ports

16x CD/DVD +/- RW drive

- 1GB PCIe video card
- PCIe sound card
- Gigabit Ethernet

Currently: PCIe 3.0/1 (1GB/s x1); PCIe 4.0 (2GB/s x1); PCIe 5.0 (4GB/s x1); graphic cards (GPUs) used as co-processors to accelerate computations

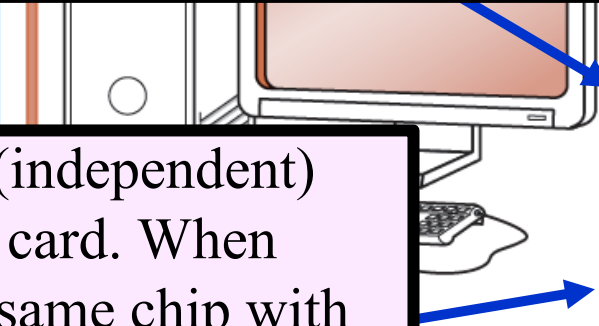


1.3.1 An Example System

In an active matrix there is one transistor per pixel.

The LCD monitor has a diagonal of 24", width:height ratio of 16:10, WUXGA resolution with 1920:1200 dots, 300 candels/m² of brightness, 1001:1 contrast ratio, 8ms response time, 24 bit colours, analog port (VGA) and digital port (DVI).

Discrete (independent) graphics card. When sharing the same chip with the CPU it is **integrated**.

- 
- 10 USB ports, 1 serial port, 4 PCI expansion slots (1 PCI, 1 PCI x 16, 2 PCI x 1), HDMI
 - 24" widescreen LCD monitor, 16:10 aspect ratio, 1920 x 1200 WUXGA, 300 cd/m², active matrix, 1000:1 (static), 8ms, 24-bit color (16.7 million colors), VGA/DVI input, 2 USB ports
 - 16x CD/DVD +/- RW drive
 - 1GB PCIe video card
 - PCIe sound card
 - Gigabit Ethernet

Currently: LCD monitors replaced CRTs; they are more economical and ergonomic; the format adopted is 16:9 or 16:10 (wide) and the dominant resolution tends to be $\geq$ FHD (1920x1080), such as 2K / 4K

1.3.1 An Example System

Various ways of connecting to data networks.

Supports wireless proximity (bluetooth) or local area (802.11) communication.

Computer

- Intel i7 Quad Core, 4.20 GHz
- 2400MHz 32GB DDR4 SDRAM
- 128KB L1 cache, 2MB L2 cache
- Dual storage (7200RPM SATA 1TB HDD, 128GB SSD)
- Wireless 802.11 + Bluetooth 4.0
- 7-in-1 card reader
- 10 USB ports, 1 serial port, 4 PCI expansion slots (1 PCI, 1 PCI x 16, 2 PCI x 1), HDMI
- 24" widescreen LCD monitor, 16:10 aspect ratio, 1920 x 1200 WUXGA, 300 cd/m², active matrix, 1000:1 (static), 8ms, 24-bit color (16.7 million colors), VGA/DVI input, 2 USB ports
- 16x CD/DVD +/- RW drive
- 1GB PCIe video card
- PCIe sound card
- Gigabit Ethernet

Supports wired network communication. The network card can be **discrete** (independent) or, if it is part of the motherboard (on-board), it is said to be **integrated**.

Currently: Bluetooth 5/6 (2/3 Mbps), Wi-Fi 6/7 (9.6/46 Gbps), 2.5 Gbps Ethernet; in servers: 10/40/100/200/400 Gbps Ethernet

1.3 Computer Components



See also:

How does Computer Hardware Work? 3D Animated Teardown

<https://www.youtube.com/watch?v=d86ws7mQYIg>

What does what in your computer? Computer parts Explained

<https://www.youtube.com/watch?v=ExxFxD4OSZ0>

Motherboards Explained

<https://www.youtube.com/watch?v=b2pd3Y6aBag>

Motherboard Form Factors

<https://www.youtube.com/watch?v=kQ1FgGidoic>

What is a Chipset?

<https://www.youtube.com/watch?v=eJn-qPHtfzI>

USB Ports, Cables, Types, & Connectors

<https://www.youtube.com/watch?v=pIZREjck9jg>

Monitors Explained - LCD, LED, OLED, CRT, TN, IPS, VA

<https://www.youtube.com/watch?v=yxygknX1AiE>

HDMI, DisplayPort, DVI, VGA, Thunderbolt - Video Port Comparison

<https://www.youtube.com/watch?v=iFO3EiQbNJ8>

Power Supplies and 80 Plus Ratings Explained

<https://www.youtube.com/watch?v=ZW1wcoERoDU>

1.4 Standards Organizations



- throughout the remainder of this subject we will see how computer parts work and how they interact with software to make a full computer

This statement raises two important questions:

- What assurance do we have that computer components will **operate as we expect?**
- And what assurance do we have that computer components **will operate together?**

1.4 Standards Organizations



- there are many organizations that define computer hardware **standards** to ensure the **interoperability** of computer components
- *Institute of Electrical and Electronic Engineers (IEEE)*
 - promotes the interests of the worldwide electrical engineering and informatics community.
 - establishes standards for computer components, data representation (**ASCII**, **floating point**), and signaling protocols, among many other things.



1.4 Standards Organizations



- *International Organization for Standardization (ISO)*



- establishes worldwide standards for everything from screw threads to photographic film.
- influential in formulating standards for computer hardware & software, including manufacture methods
- examples: **C95, C99, C11, C17** (ISO/IEC 9899:2018)

Note: ISO is **not** an acronym;
it comes from the Greek *isos* meaning “equal”

1.4 Standards Organizations

- *International Telecommunications Union (ITU)*
 - defines norms for telecommunications systems (data communications, telephony) and IT
 - example: **X.509** (certificates)
 - under the supervision of the United Nations
- some national groups establish standards within their respective countries:
 - American National Standards Institute (ANSI)
 - example: **ANSI C** (C85, C96, C89)
 - British Standards Institution (BSI)



1.5 Historical Development



- to fully appreciate the computers of today, it helps to understand how things got the way they are
- the evolution of computing machinery has taken place over several centuries
- in modern times, computer evolution is usually classified into 4 generations according to the fundamental underlying technology of the era

See also:

The History of Computing

<https://www.youtube.com/watch?v=-M6lANfzFsM>

The Incredible Evolution of Computers

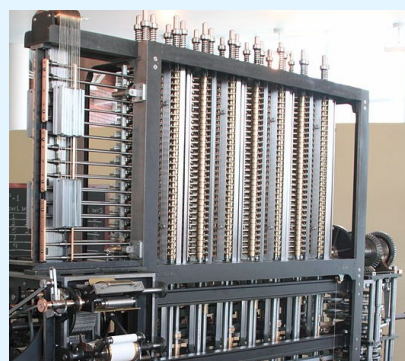
<https://www.youtube.com/watch?v=aa6YISbAJEA>

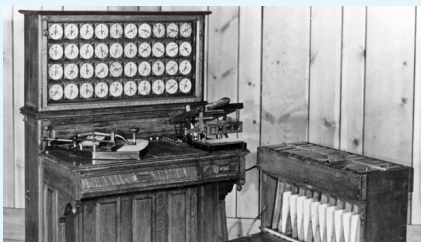
Evolution of Computers (1623–2100): Full Timeline

<https://www.youtube.com/watch?v=c-DiBQYGV-g>

1.5 Historical Development



- Gen. 0: **Mech. Calculating Machines** (1642 - 1945)
 - **Calculating Clock:**
Wilhelm Schickard (1592 - 1635) → 
 - **Pascaline:**
Blaise Pascal (1623 - 1662) → 
 - **Difference Engine:**
Charles Babbage (1791 – 1871) → 
 - **Punched card tabulating machines:**
Herman Hollerith (1860 - 1929).



Hollerith cards were commonly used for computer input well into the 1970s.

1.5 Historical Development



- Gen. 1: **Vacuum Tube Computers** (1945 - 1953)

- the computer

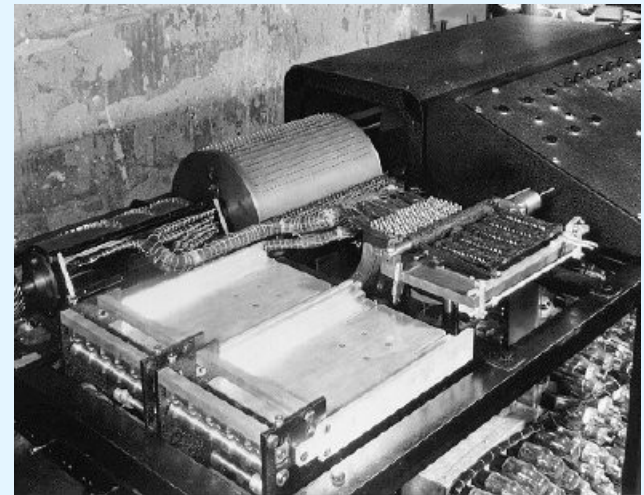
[Atanasoff Berry](#)

(1937-1938) solved linear equations

- John Atanasoff & Clifford Berry, of Univers. of Iowa



vacuum tubes

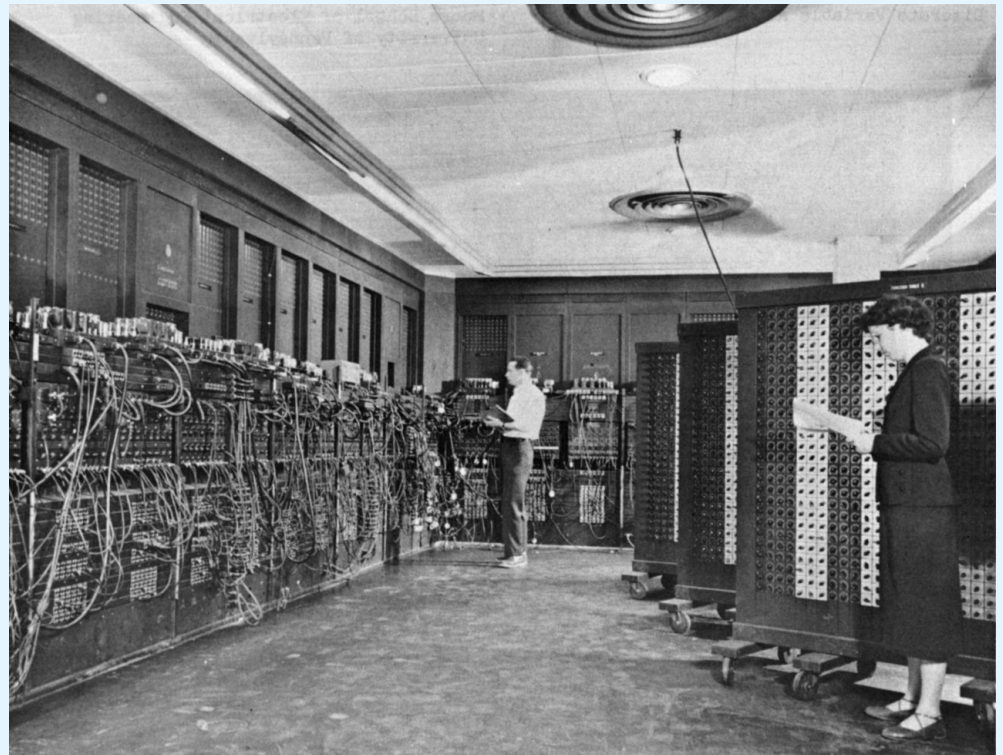


- the 1st computer i) built on non-mechanical parts, ii) that used the binary system to represent data, iii) that separated the computational part from the storage part

1.5 Historical Development



- Gen. 1 (cont.)
 - Electronic Numerical Integrator and Computer (**ENIAC**)
 - The first **generic** use computer
 - John Mauchly & J. Presper Eckert, University of Pennsylvania, 1946



1.5 Historical Development

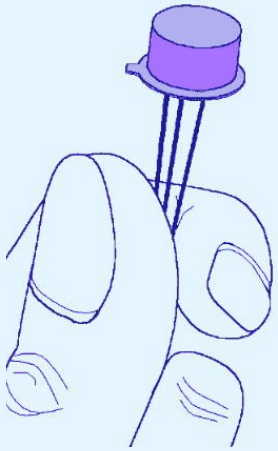
- Gen. 1 (cont.)



- **IBM 650** (1955, phased out in 1969)
 - the first **mass produced** computer

1.5 Historical Development

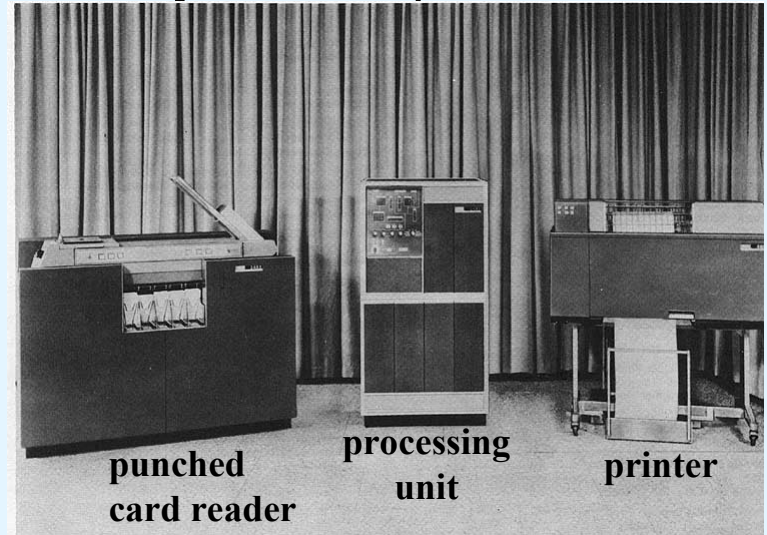
- Gen. 2: **1st Transistorized Computers (1954 - 1965)**



- IBM **7094** (scientific)
and **1401** (business)

- Digital Equipment Corporation (DEC)
PDP-1

- **Univac 1100** ...
... and many others.



These systems had few architectural similarities.

DEC PDP-1

1.5 Historical Development

- Gen. 3: **Integrated Circuit Computers** (1965-1980)

- **IBM 360**, based on SLT circuits
- **DEC PDP-11**
(the most popular mini-computer)
- super computer **Cray-1**
- ... and many others



1.5 Historical Development



- Gen. 4: **VLSI Computers** (1980 - ????)
 - *VLSI circuits* = *very large scale integrated circuits*; > 10 000 components in a single circuit
 - enabled the creation of **microprocessors**
 - the 1st was the **Intel 4004** (4 bits); next versions (8080/86/88) helped to propagate the “**personal computer**”
 - today, integrated circuits include billions of transistors, with distances between them of just a few *nm*



Intel 4004

1.5 Historical Development



- **Moore's Law (1965)**
 - Gordon Moore, Intel founder
 - Moore observed an industry trend: “The density of transistors in a integrated circuit **doubled every year.**” (later, that period was revised to 18 and 24 months)
 - but this trend slowed down some time ago, due to the **increasing costs** and also **limitations from Physics**
 - thus, it is known, for some time, that Moore's Law is no longer valid (<https://towardsdatascience.com/moores-law-is-dead-678119754571>)
 - meanwhile, other models of computing are emerging (e.g. quantic computing) as possible future alternatives

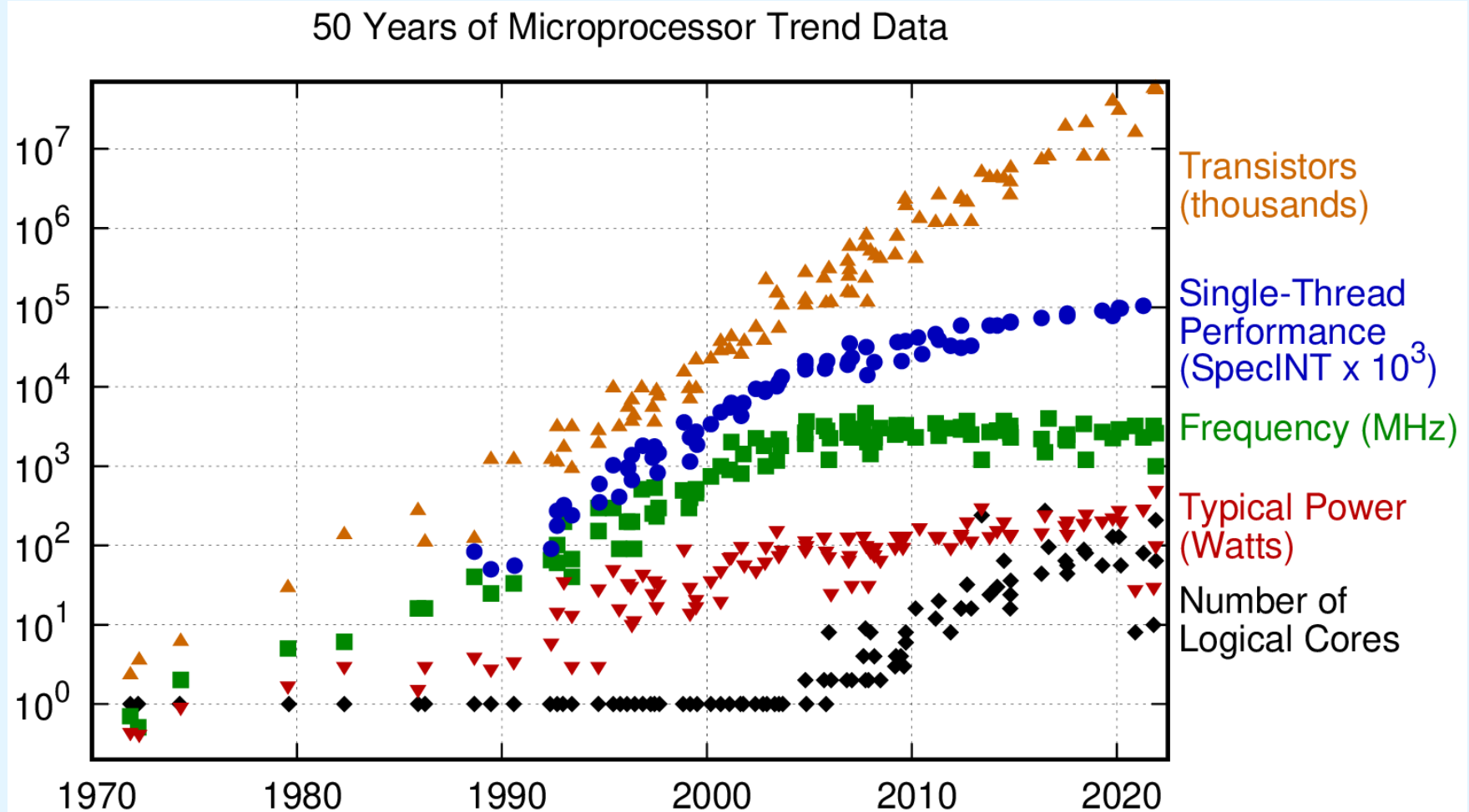
1.5 Historical Development

- **Dennards's Law / Dennard's Scaling (1974)**
 - Robert H. Dennard, IBM (1974); as transistors shrink, power density remains constant, once voltage and current scale down with size (*)
 - allowed faster, more efficient chips without exceeding thermal limits, enabling the exponential performance gains of the Moore's Law era (**)
 - this principle broke down around 2005-2006, when reducing voltage further lead to leakage current, causing chips to exceed thermal limits
 - the end of this scaling led to "dark silicon" (parts of the chip must be powered off to manage heat), the rise of multi-core processors (instead of higher clock speeds), and a focus on power-efficient architectures

(*) reducing transistor dimensions by a factor of S , the circuit delay decreases by S (running faster), and power consumption decreases by S^2 , keeping power density constant

(**) allowed for roughly twice the number of transistors per chip, at higher speeds, with the same power budget

1.5 Historical Development



1.5 Historical Development



- **Rock's Law** (aka 2nds Moore's Law)
 - Arthur Rock, Intel financier;
 - observed that “The cost of capital equipment to build semiconductors will double every four years.”
 - in 1968, a new chip plant cost about \$12 000 (the price of a nice home in the suburbs); but these costs increased considerably: > 2 billion dollars in 2003, 5 billions in 2012, 14 billions in 2015 (which is > than the gross domestic product (GDP) of some countries)
 - but the rate of growing has been decreasing due to increased efficiencies in the production process

1.5 Historical Development



See also:

checkpoint:
exercise 1.7

The End of Moore's Law?!

<https://www.youtube.com/watch?v=xnGWRReXRlk>

Why Moore's Law Matters

<https://www.youtube.com/watch?v=nRJgvX6P8dI>

The End of Dennard Scaling

<https://www.youtube.com/watch?v=7p8ZeSbblec>

Why 10GHz Silicon CPUs Are Impossible

https://www.youtube.com/watch?v=5JWcI_xutuI

1.6 The Computer Level Hierarchy



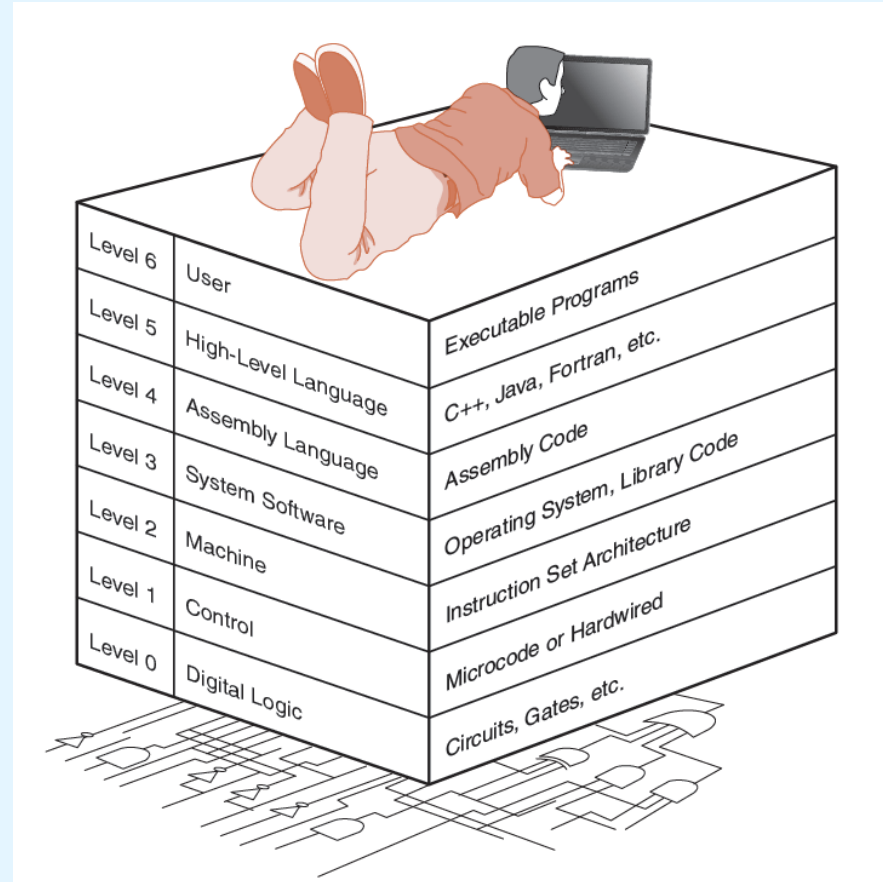
- computers consist of many things besides chips
- true, ultimately it will be the physical components (the so-called **Hardware**) that will carry out the tasks
- but for this to happen, the Computer must support the description of a task through various types of logical representation (the so-called **Software**), and the conversion between these types and to the Hardware
- thus, between the world of Software and the lowest level of Hardware there is a *semantic gap*

1.6 The Computer Level Hierarchy

- to help bridge this semantic gap, and make it easier to implement Computers and understand how they work, a Computer is organized into a **hierarchy of layers or levels**
 - each layer performs a well-defined function, and provides a normalized interface to the layer(s) above
- next, a simplified hierarchy is presented

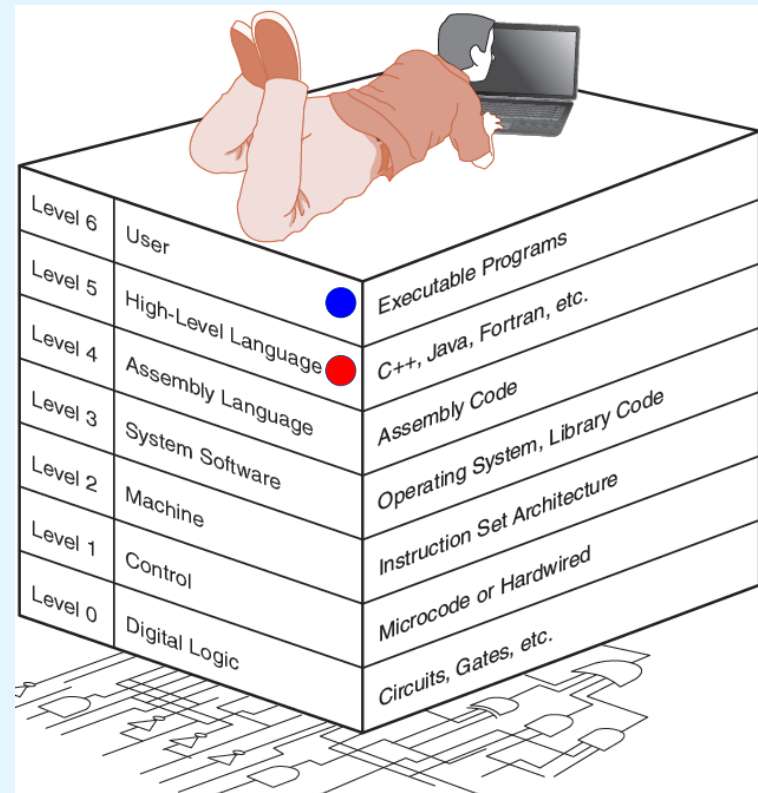
1.6 The Computer Level Hierarchy

- each layer can be seen as a virtual machine, abstracting the level/machine below it
- the machines at each level execute their own particular instructions, calling upon machines at lower levels to perform tasks as required
- ultimately, the computer circuits carry out the work



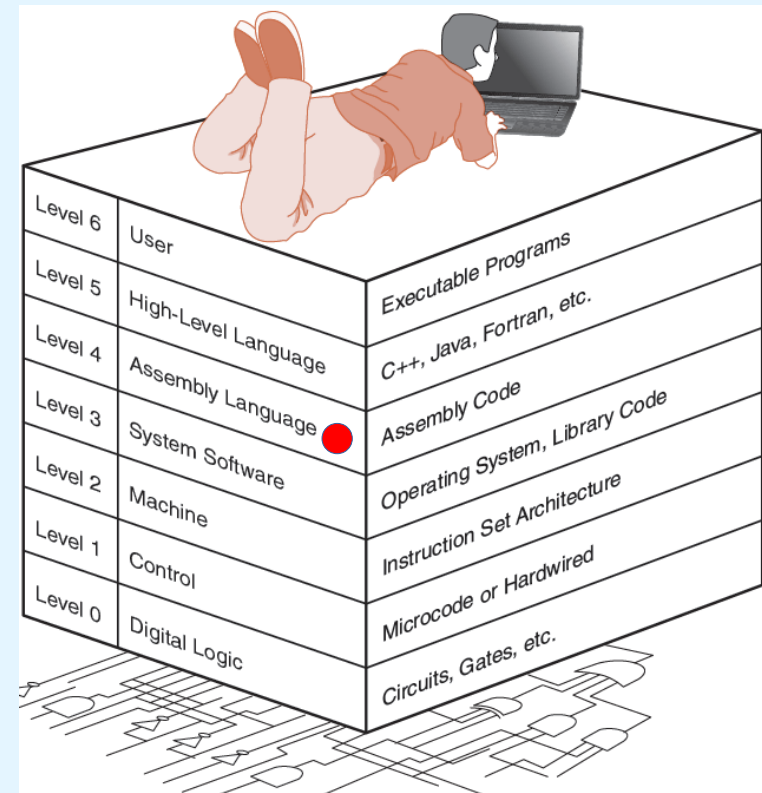
1.6 The Computer Level Hierarchy

- level 6: **User**
 - program execution and user interface level
 - the level with which we are most familiar
- level 5: **High-Level Languages**
 - level with which we interact when we write programs in languages such as C, Java, ...
 - level familiar to programmers



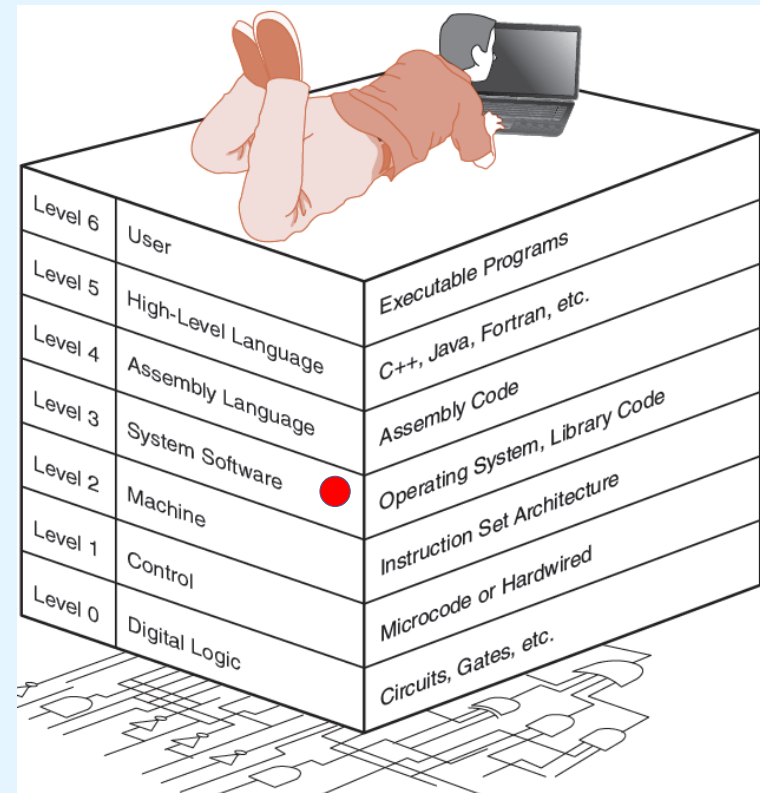
1.6 The Computer Level Hierarchy

- level 4: **Assembly Language**
 - deals with i) assembly instructions produced in Level 5 by the compilation of high-level programs, or ii) programs written directly in assembly language



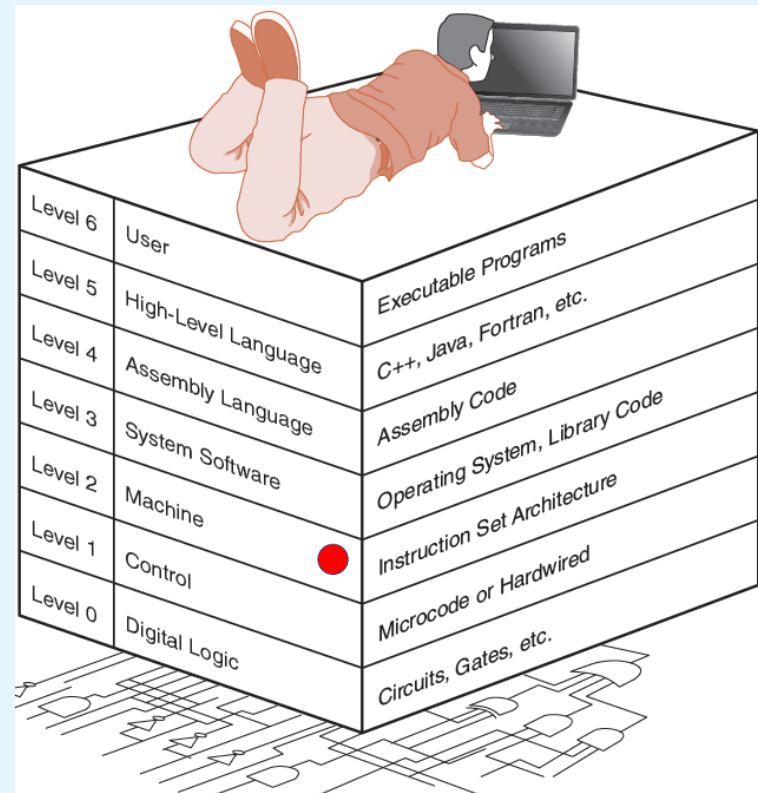
1.6 The Computer Level Hierarchy

- level 3: **System Software**
 - Operating System level
 - controls the execution of programs on the system
 - protects system resources
 - assembly language instructions often pass through Level 3, to Level 2, without modification



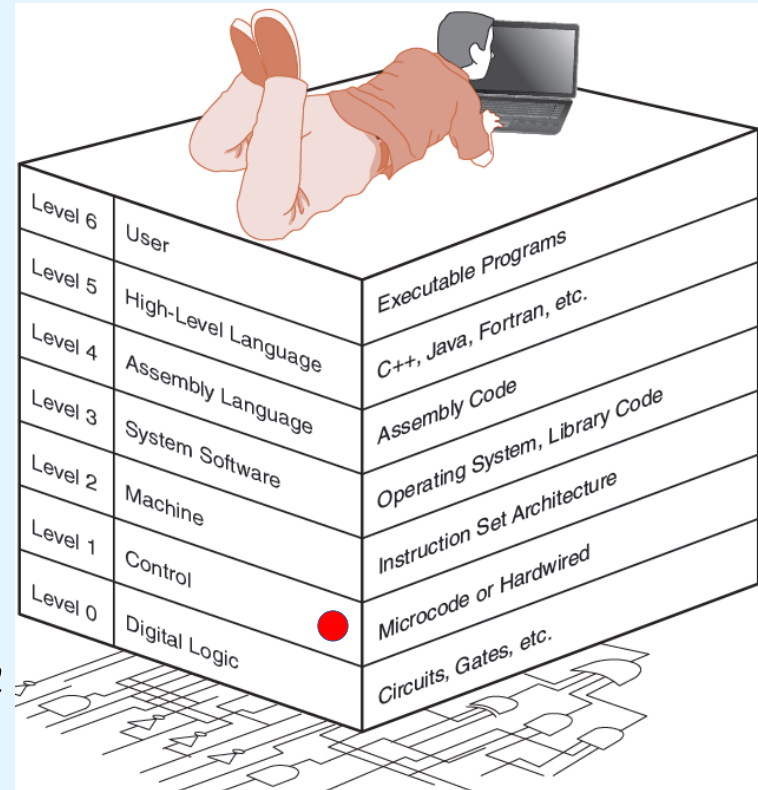
1.6 The Computer Level Hierarchy

- level 2: **Machine Level**
 - also known as the Instruction Set Architecture (ISA) level
 - consists of binary instructions, particular to the architecture of the machine
 - programs written in machine language need no compilers, interpreters, or assemblers (the later directly translate programs in textual assembly to numerical machine code)



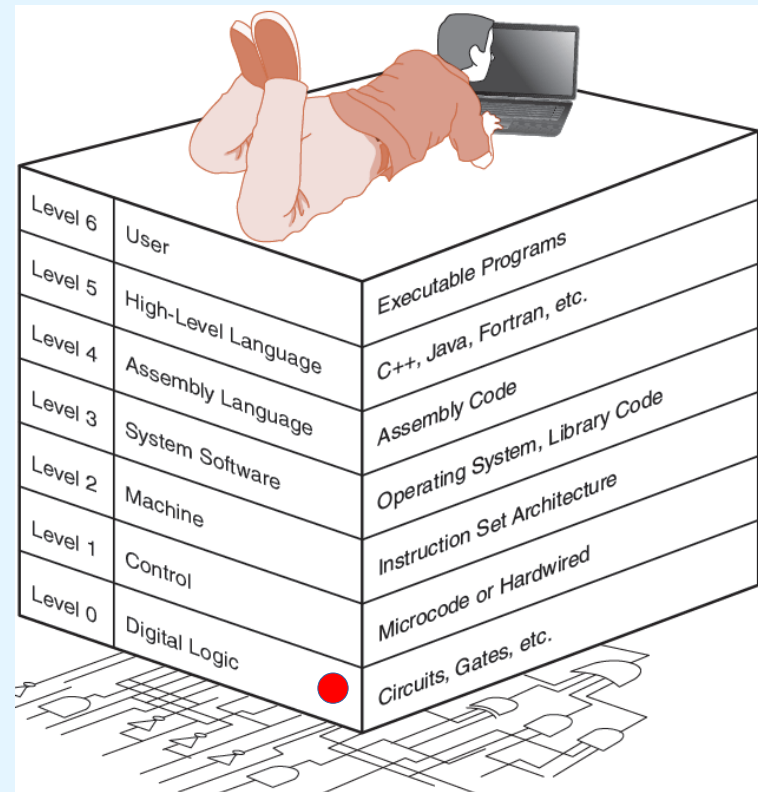
1.6 The Computer Level Hierarchy

- level 1: **Control**
 - a *control unit* decodes and executes instructions and moves data through the system
 - control units can be A) *micro-programmed* or B) *hardwired*
 - A: each machine-instruction is implemented by a *microprogram* written in a low-level language; each *microinstruction* is implemented directly in hardware
 - B: each machine-instruction is *directly implemented in hardware* (no further translations are needed)



1.6 The Computer Level Hierarchy

- Level 0: **Digital Logic**
 - level where the digital circuits (the chips) sit
 - digital circuits consist of *gates* and *wires*
 - they implement the mathematical logic that sustains all other levels

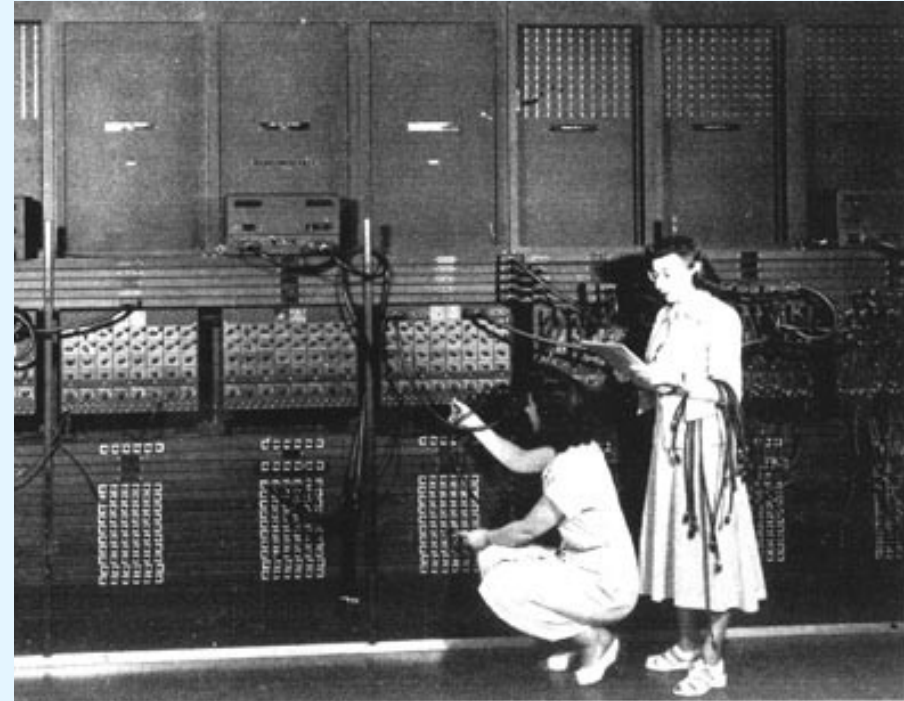


Complemento: HOW TRANSISTORS RUN CODE? [Core Dumped]

<https://www.youtube.com/watch?v=HjneAhCy2N4&list=PL9vTTBa7QaQOoMfpP3zvtvgyQkPWDPfJez>

1.7 The von Neumann Model

- on the ENIAC, all programming was done at the **digital logic** level
- programming the computer involved moving plugs & wires
- a different hardware configuration was needed to solve every unique problem type !



Configuring the ENIAC to solve a “simple” problem required many days labor by skilled technicians.

1.7 The von Neumann Model



- but, the inventors of the ENIAC, John Mauchley and J. Presper Eckert, conceived of a computer (EDVAC) that could **store instructions in memory**
- however, this invention has since been ascribed to a mathematician, John von Neumann, who was a contemporary of Mauchley and Eckert
- since then, stored-program computers become known as **von Neumann Architecture** systems

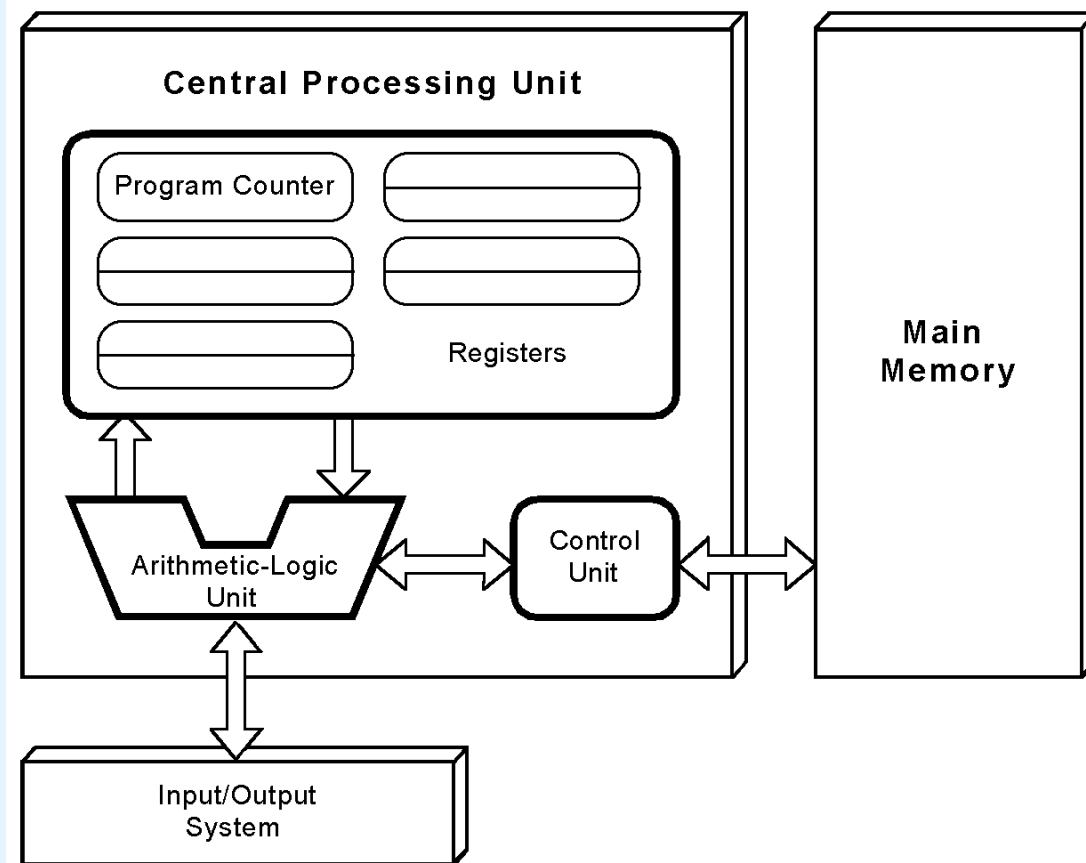
1.7 The von Neumann Model



- nowadays, stored-program computers have the following minimal set of characteristics:
 - three hardware systems:
 - a central processing unit (CPU)
 - a main memory system
 - an I/O system
 - capacity to carry out sequential instruction processing
 - a single data path between the CPU and main memory
 - aka the *von Neumann bottleneck*

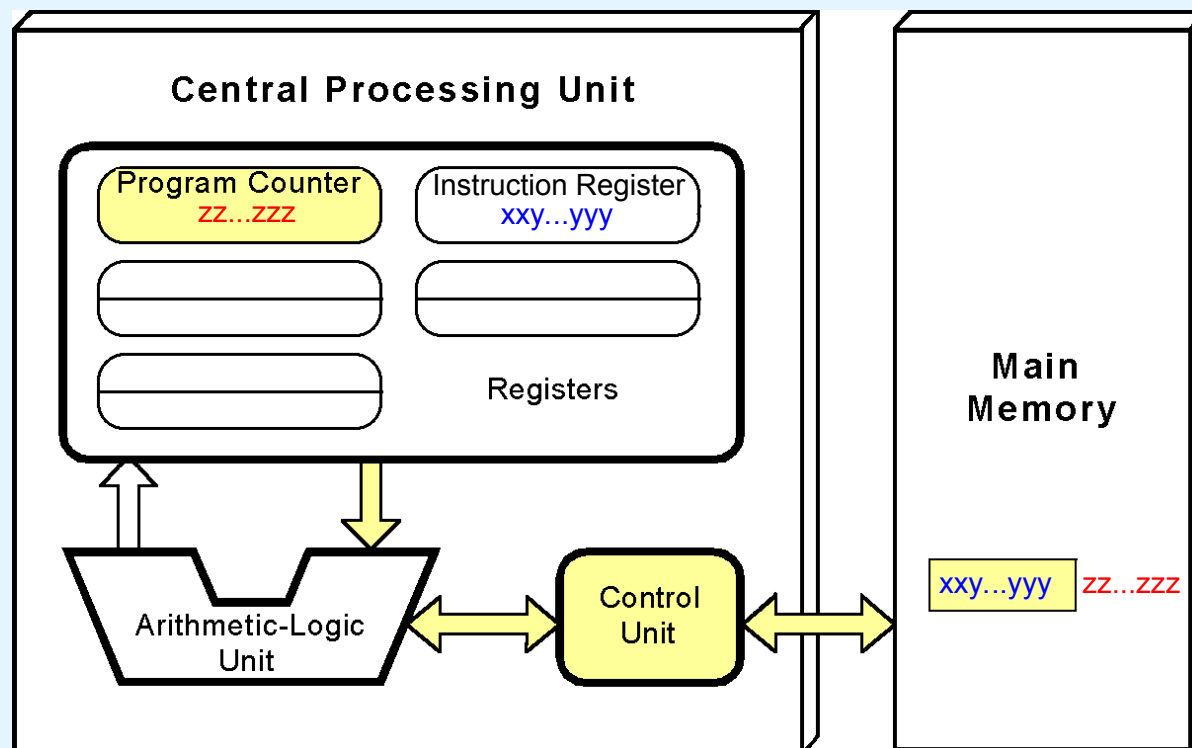
1.7 The von Neumann Model

- this is a general depiction of a von Neumann system →
- it follows a **fetch-decode-execute** cycle to run programs



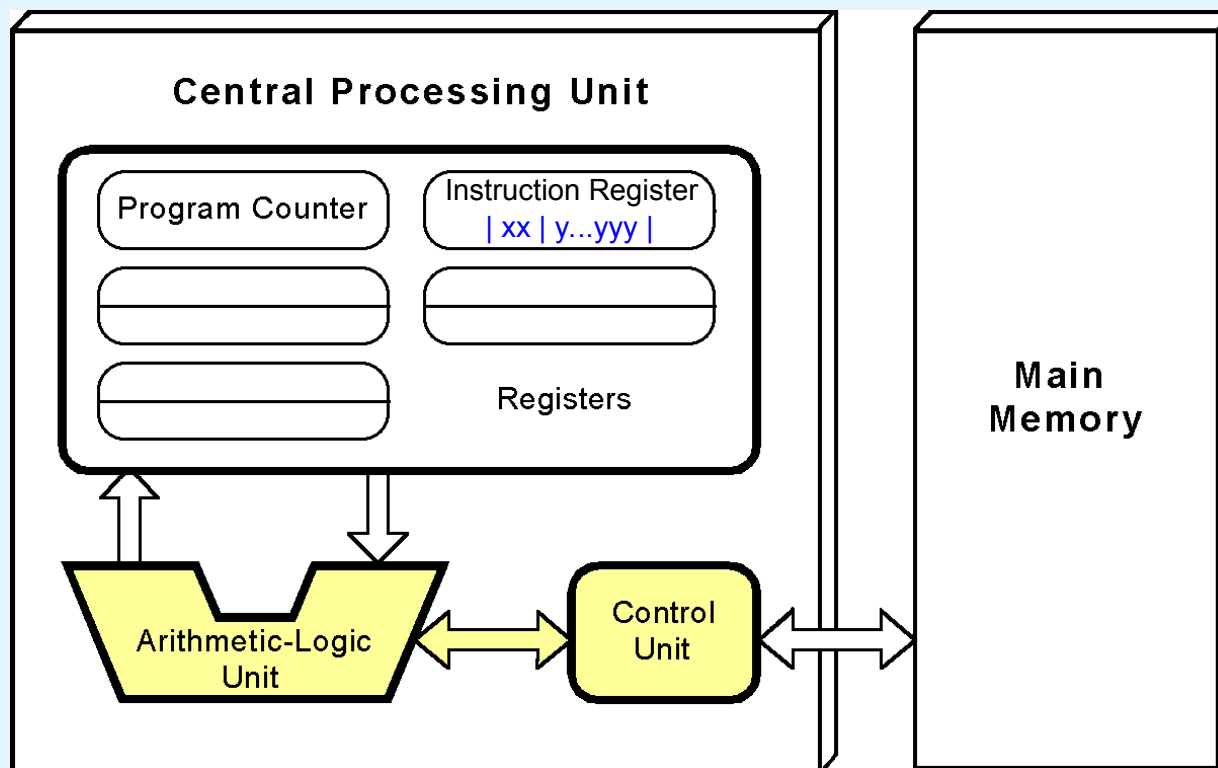
1.7 The von Neumann Model

- **fetch:** the *Control Unit* reads the *next instruction* from *memory*, using the **PC (Program Counter)** register to determine where (at which *memory address*) the instruction is located
 - the instruction is placed in the **IR (Instruction Register)**



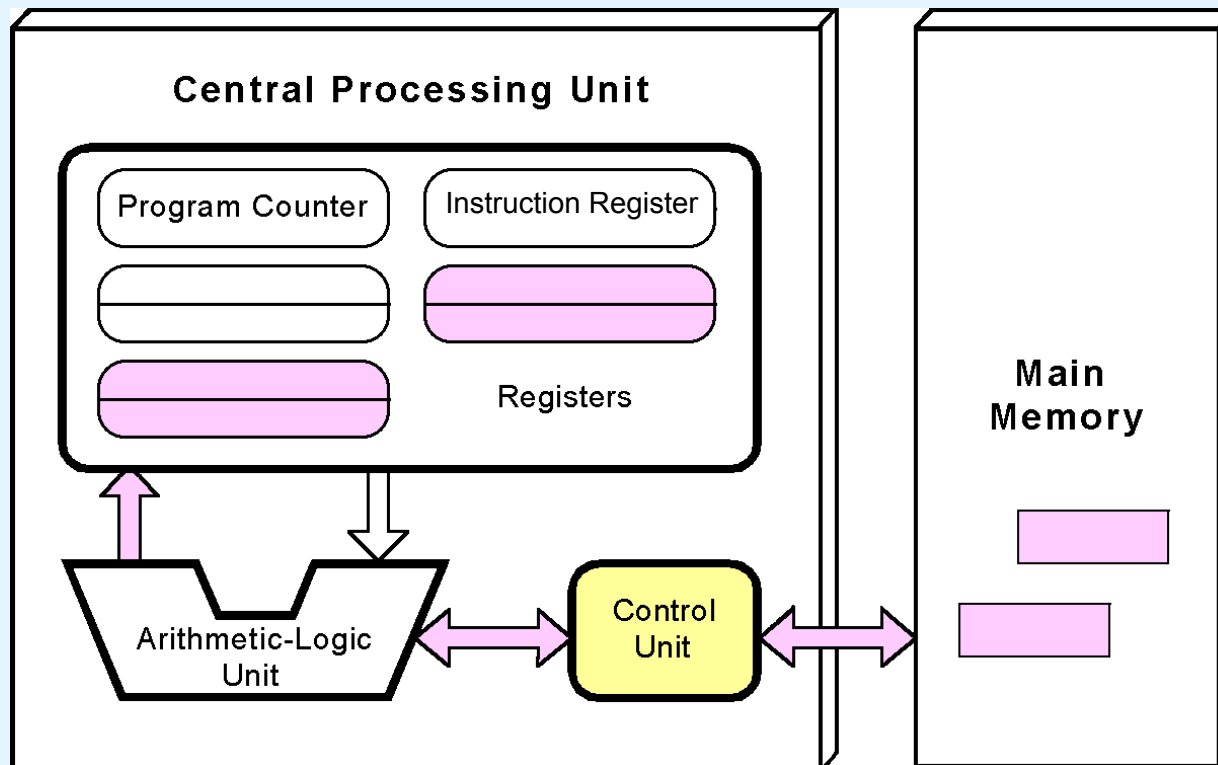
1.7 The von Neumann Model

- **decode:** the *instruction* in the IR is decoded into a language that the ALU (Arithmetic Logic Unit) can understand



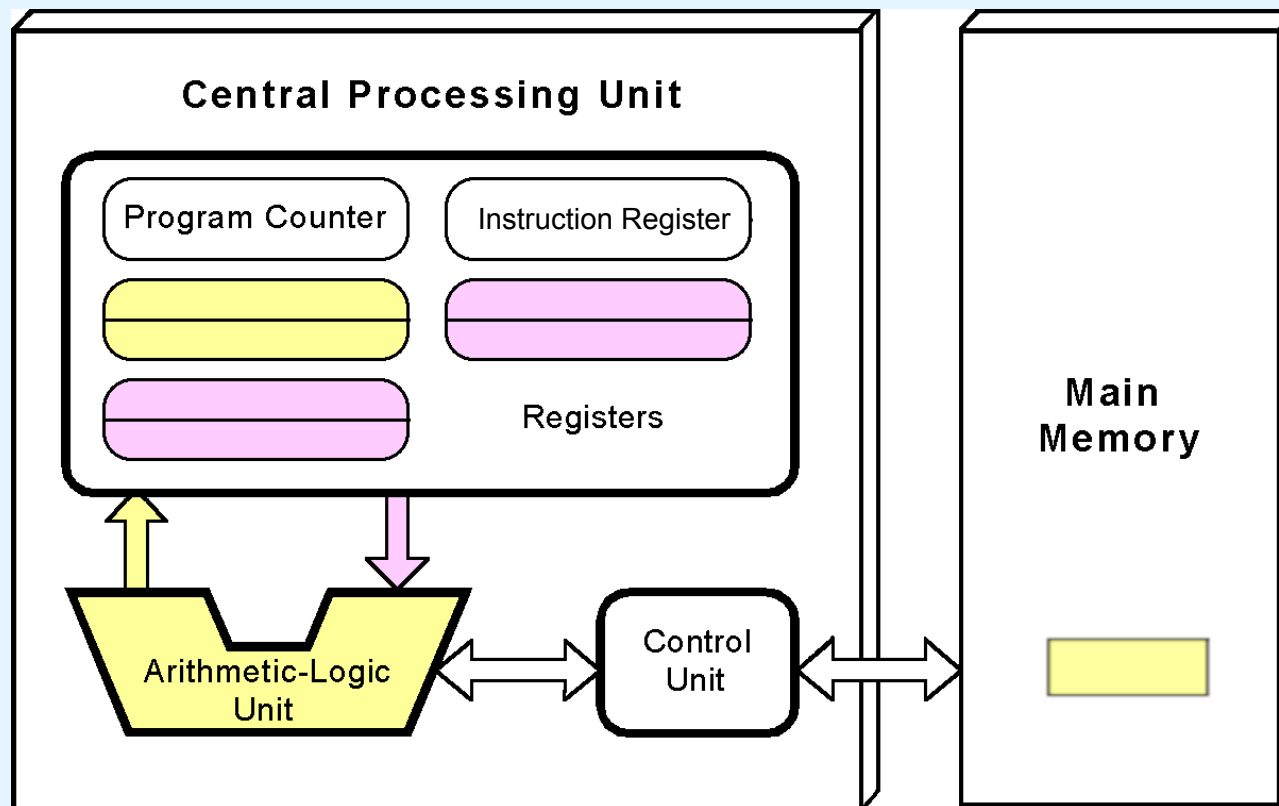
1.7 The von Neumann Model

- **decode**: any data **operands** required to execute the instruction are fetched from memory and placed into CPU registers



1.7 The von Neumann Model

- **execute**: the ALU executes the instruction and places the **results** of that execution in registers or memory



1.7 The von Neumann Model



See also:

How Do CPUs Work? [Explaining Computers]

<https://www.youtube.com/watch?v=jx-w2o-Lj8g>

CPU Architecture Explained [Bit Lemon]

<https://www.youtube.com/watch?v=GtVDTp826DE>

How do CPUs Work? [Branch Education]

<https://www.youtube.com/watch?v=16zrEPOsIcI>

1.8 Non-von Neumann Models



- conventional stored-program computers have undergone many incremental improvements over the years
 - specialized buses (e.g., PCI), floating-point units, pipelining, cache memories, to name only a few.
- enormous improvements in computational power require departure from the classic von Neumann architecture
 - separated buses for instructions and data (Harvard architecture)
 - multi-processor systems, and their interconnection (clusters)
 - mixing processors of different architectures (heterogeneous systems: CPUs + GPUs)

1.8 Non-von Neumann Models



- parallel / distributed / heterogeneous processing have been providing increased computational power for many years now, allowing to compensate for the difficulty in sustaining Moore's Law validity
- more radical approaches are also being investigated
 - quantum computing, photonic systems
 - but their effective viability is still uncertain

See also: Quantum Computing Explained
<https://www.youtube.com/watch?v=WiIX2oNldhk>

Conclusion



- This unit provided an overview of the subject of computer architecture and related topics.
- The general structure of a computer system was presented, laying the foundations for more advanced topics to be next introduced.
- Subsequent units will explore many of these topics in great detail (and other topics).

References



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